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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
Negative training images were extracted at random positions from the tomograms, but the volume of the actin bundle was excluded.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
In the input starfile, the refined psi angles and translations found in the previous step were passed as priors plus for each segment a random rot angle as prior was added (rlnAngleRotPrior) and the tilt prior (rlnAngleTiltPrior) was set to 90deg (He and Scheres, 2017).
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.
Cell Line Authentication

Authentication: Furthermore, CTF correction in RELION jobs was switched off, because the tomograms were CTF corrected by phase-flipping prior to subtomogram extraction.
Code Information
Availability: Data and code availability The APT code is available at github.commeibauerActinPolarityToolbox.
Data Information
Availability: The manual segmented dataset (7 cryo-tomograms of FAs) together with the APT workflow is deposited at the Electron Microscopy Public Image Archive under accession code EMPIAR-10570.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Deposited data Manually segmented dataset (7 cryo-tomograms of FAs from MEFs) and APT workflow This paper EMPIAR-10570 In-situ structure of an actin filament from FAs of MEFs This paper EMD-11976 Near-atomic resolution for one state of F-actin (Galkin et al., 2015) EMD-6179 Talin-activated vinculin interacts with branched actin networks to initiate bundles (Boujemaa-Paterski et al., 2020) EMD-10737 Crystal-structure of the 20s proteasome from the archaeon T-acidophilum at 3.4-angstrom resolution (Lowe et al., 1995) PDB-1PMA Crystal structure of the thermosome, the archaeal chaperonin and homolog of CCT (Ditzel et al., 1998) PDB-1A6D Experimental models: cell lines MEFs stably expressing vinculin-venus (Ringer et al., 2017) NA Software and algorithms Actin polarity toolbox (APT) This paper github.commeibauerActinPolarityToolbox RELION (Scheres, 2012) https:www3.mrc-lmb.cam.ac.ukrelionindex.phpMain_Page MATLAB MathWorks, Natick, USA NA SerialEM (Mastronarde, 2005) https:bio3d.colorado.eduSerialEM MotionCorr (Li et al., 2013) https:cryoem.ucsf.edusoftware TOM toolbox (Nickell et al., 2005) https:www.biochem.mpg.de6348566tom_e AMIRA Thermo Fisher Scientific, Waltham, USA NA EMAN2 (Chen et al., 2017) https:blake.bcm.eduemanwikiEMAN2 UCSF chimera (Pettersen et al., 2004) https:www.cgl.ucsf.educhimera Resource availability Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Ohad Medalia, omedaliabioc.uzh.ch. Materials	MATLAB		Suggestion: MATLAB, RRID:SCR_001622(link)

availability This study did not generate new unique reagents.			
The manual segmented dataset (7 cryo-tomograms of FAs) together with the APT workflow is deposited at the Electron Microscopy Public Image Archive under accession code EMPIAR-10570.	Image Archive		
First, a starfile was created only passing the name of each projected subtomogram (rlnImageName) and its originating tomogram (rlnMicrographName) to RELION.	RELION		Suggestion: RELION (RRID:SCR_016274)(link)

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq